

Lab 4: Subnetting and Supernetting Using Cisco Packet Tracer

Objectives

- To understand the concepts of subnetting and supernetting.
- To design a network topology using Cisco Packet Tracer.
- To configure and implement subnetted and supernetted networks.
- To verify network connectivity and routing efficiency.

Software and Hardware Requirements

- Cisco Packet Tracer
- Windows Operating System
- Computer/Laptop
- Keyboard and Mouse

Theory

Subnetting

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets. Instead of placing all devices in a single network, subnetting splits the network into smaller parts, where each subnet works as an independent network but remains part of the main network.

Subnetting is used to improve network performance, security, and management. In a large network, excessive broadcast traffic can slow down communication. By creating smaller subnets, broadcast traffic is reduced, security is enhanced, and network administration becomes easier.

A subnet mask is used to separate the network portion from the host portion of an IP address. It consists of 1s for network bits and 0s for host bits. For example, the subnet mask 255.255.255.0 indicates that the first 24 bits are for the network and the remaining 8 bits are for hosts.

Subnetting works by borrowing bits from the host portion of an IP address to create additional subnets. Borrowing more bits increases the number of subnets but reduces the number of hosts per subnet. This trade-off allows networks to be designed according to requirements.

Number of subnets = 2^n , where n = borrowed bits

Number of hosts per subnet = $2^h - 2$, where h = remaining host bits

(The subtraction of 2 is for the network and broadcast addresses.)

Each subnet has:

A network address (first address)

A broadcast address (last address)

These addresses cannot be assigned to hosts.

Supernetting

Supernetting is the process of combining multiple smaller, contiguous IP networks into a single larger network called a supernet. It is the opposite of subnetting.

Supernetting is mainly used to reduce routing table size. When routers store many individual routes, routing becomes slow and inefficient. By summarizing multiple networks into a single route, routing performance improves.

In supernetting, fewer bits are used for the network portion, creating a larger address block. As a result, multiple networks are advertised as one summarized route.

Supernetting is widely used by Internet Service Providers (ISPs). ISPs allocate smaller address blocks to customers but advertise them as a single summarized route to the internet. This improves routing efficiency, reduces memory usage in routers, and simplifies network management. However, supernetting must be done carefully, as combining non-contiguous networks can cause routing problems.

Subnetting

Network Design:

Base Network: 192.168.1.0/24

Required number of subnets: 4

Number of IP Address per subnets: 64

Number of host bits: 6 bits

Subnet	Network ID	Broadcast ID	First Host	Last Host
1	192.168.1.0/26	192.168.1.63/26	192.168.1.1/26	192.168.1.62/26
2	192.168.1.64/26	192.168.1.128/26	192.168.1.65/26	192.168.1.126/26
3	192.168.1.128/26	192.168.1.191/26	192.168.1.129/26	192.168.1.190/26
4	192.168.1.192/26	192.168.1.255/26	192.168.1.193	192.168.1.254/26

Configuration:

Device	Interface / Subnet	IP Address	Subnet Mask	Default Gateway
Router0	FastEthernet0/0	192.168.1.1	255.255.255.192	N/A
Router0	FastEthernet0/1	192.168.1.65	255.255.255.192	N/A
PC0	Subnet 1	192.168.1.2	255.255.255.192	192.168.1.1
PC1	Subnet 1	192.168.1.3	255.255.255.192	192.168.1.1
PC2	Subnet 1	192.168.1.4	255.255.255.192	192.168.1.1
PC3	Subnet 2	192.168.1.66	255.255.255.192	192.168.1.65
PC4	Subnet 2	192.168.1.67	255.255.255.192	192.168.1.65
PC5	Subnet 2	192.168.1.68	255.255.255.192	192.168.1.65

Network Topology:

PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

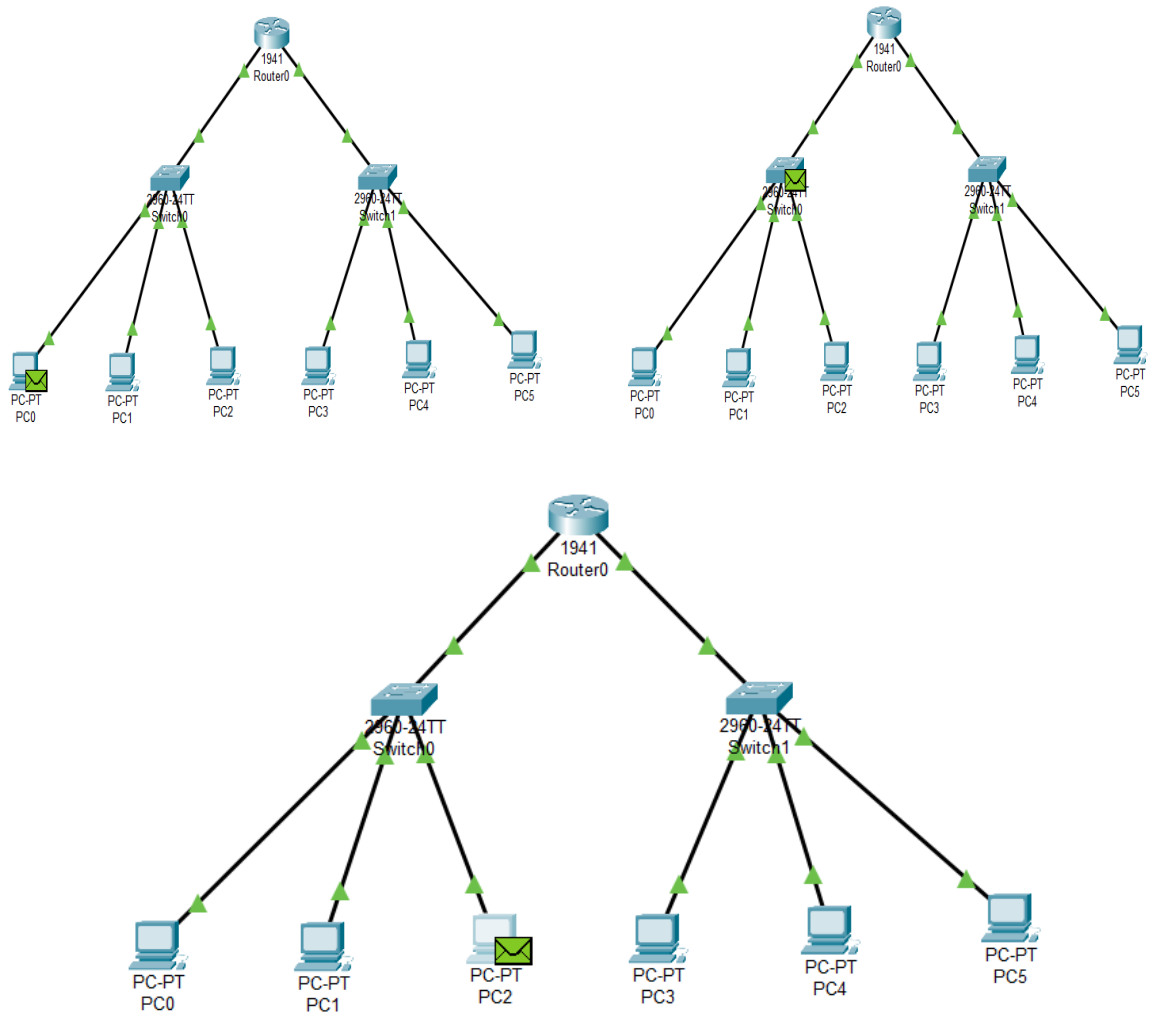
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=8ms TTL=128
Reply from 192.168.1.4: bytes=32 time=4ms TTL=128
Reply from 192.168.1.4: bytes=32 time=4ms TTL=128
Reply from 192.168.1.4: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms

C:\>|
```



PC1

Physical Config Desktop Programming Attributes

Command Prompt

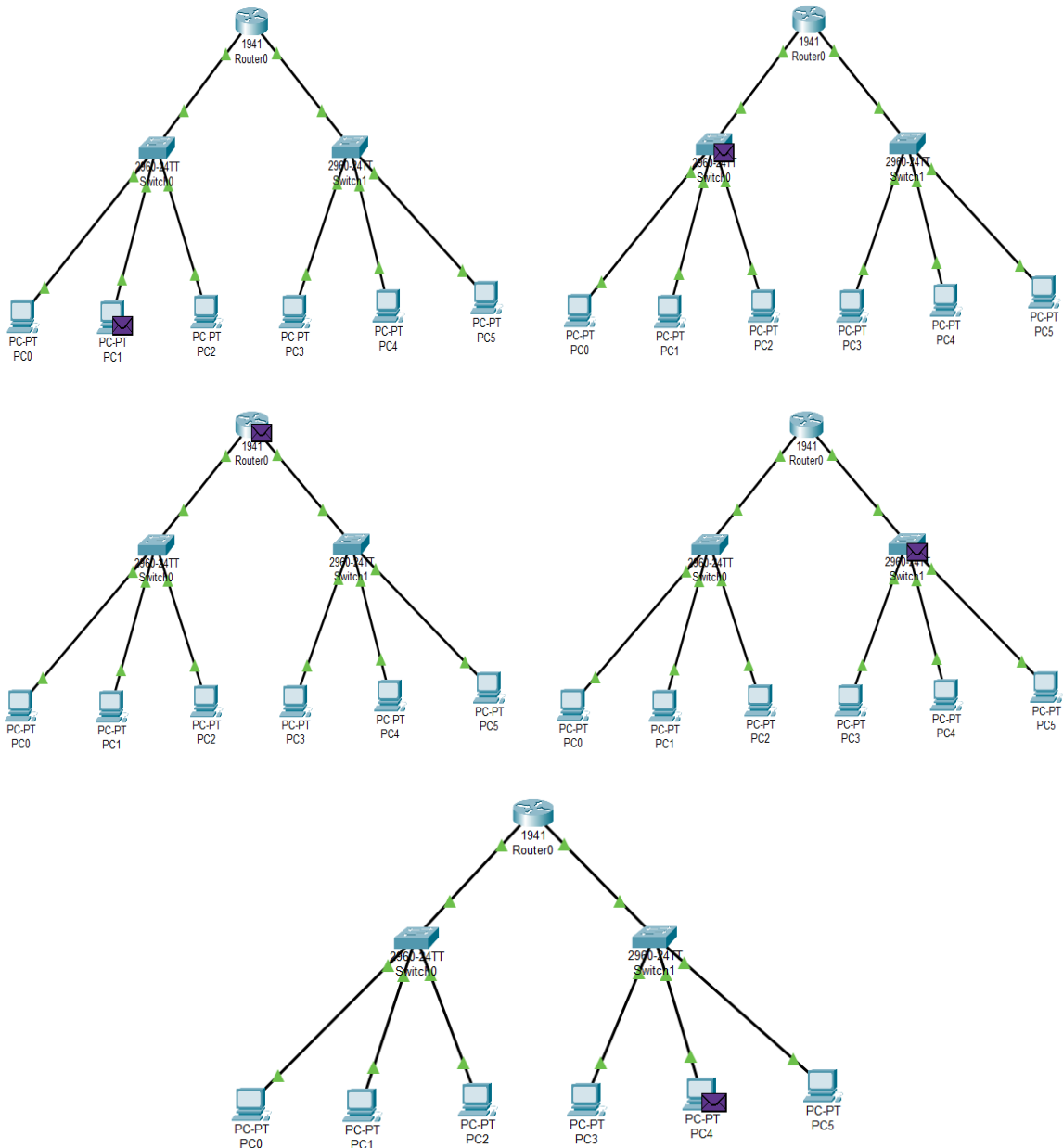
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.67

Pinging 192.168.1.67 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.67: bytes=32 time=8ms TTL=127
Reply from 192.168.1.67: bytes=32 time=8ms TTL=127
Reply from 192.168.1.67: bytes=32 time=8ms TTL=127

Ping statistics for 192.168.1.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 8ms, Average = 8ms

C:\>|
```



Supernetting

Configuration:

Given networks,

192.168.0.0/24

192.168.1.0/24

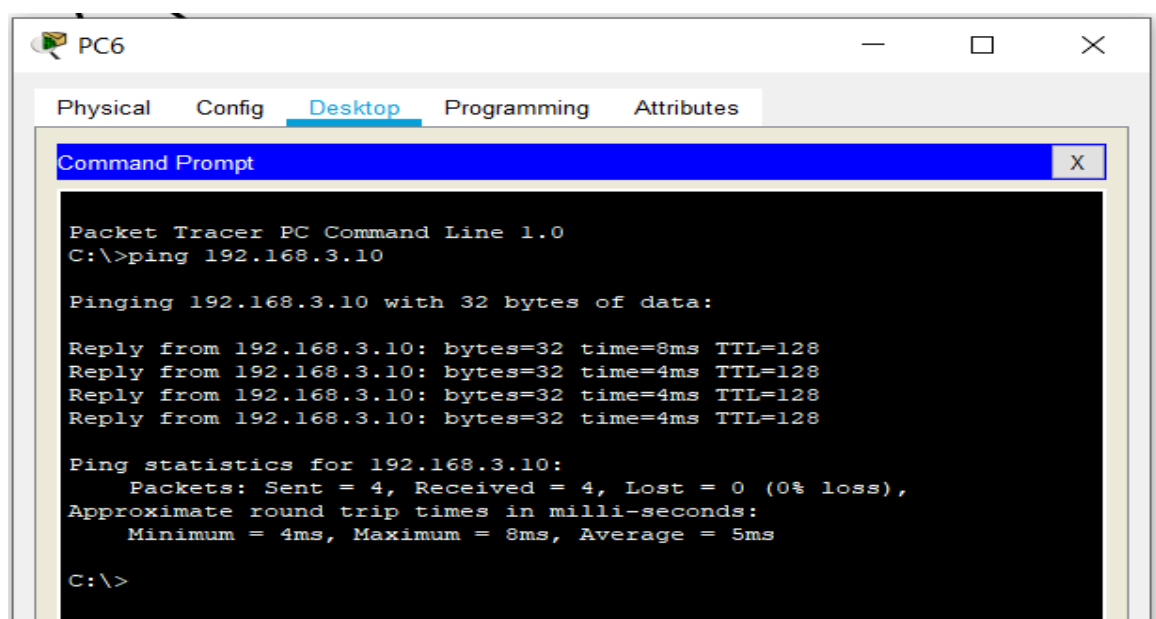
192.168.2.0/24

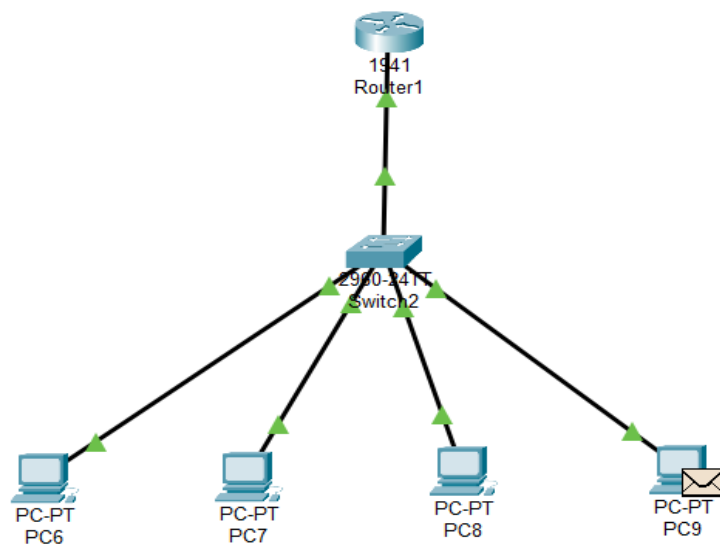
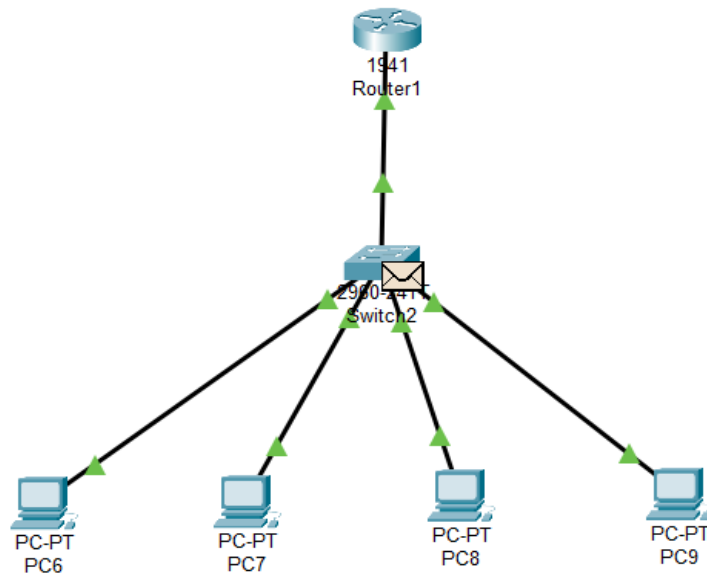
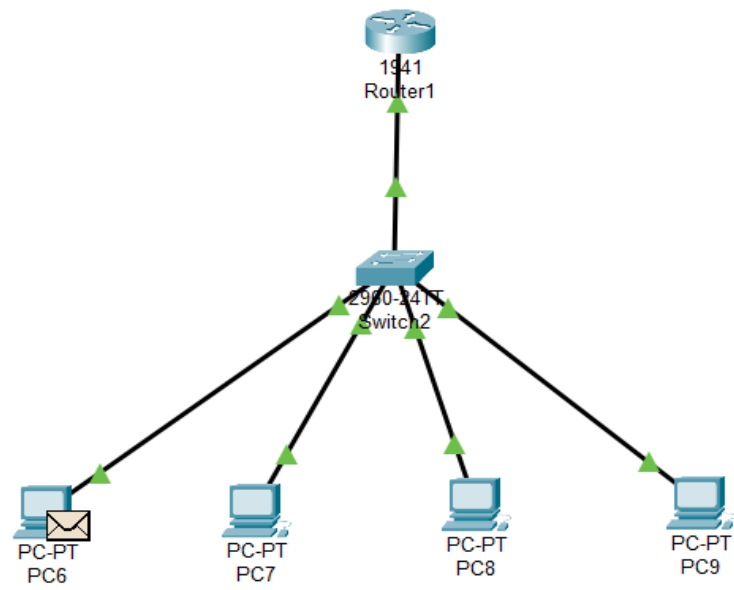
192.168.3.0/24

Subnet Address: 192.168.0.0/22

Device	IPv4 Address	Subnet Mask	Default Gateway
Router0 (FastEthernet0/0)	192.168.0.1	255.255.252.0	N/A
PC6	192.168.0.10	255.255.252.0	192.168.0.1
PC7	192.168.1.10	255.255.252.0	192.168.0.1
PC8	192.168.2.10	255.255.252.0	192.168.0.1
PC9	192.168.3.10	255.255.252.0	192.168.0.1

Network Topology:





Assignment: Subnetting

Network Design:

Base Network: 200.100.10.0/24

Required number of subnets: 2

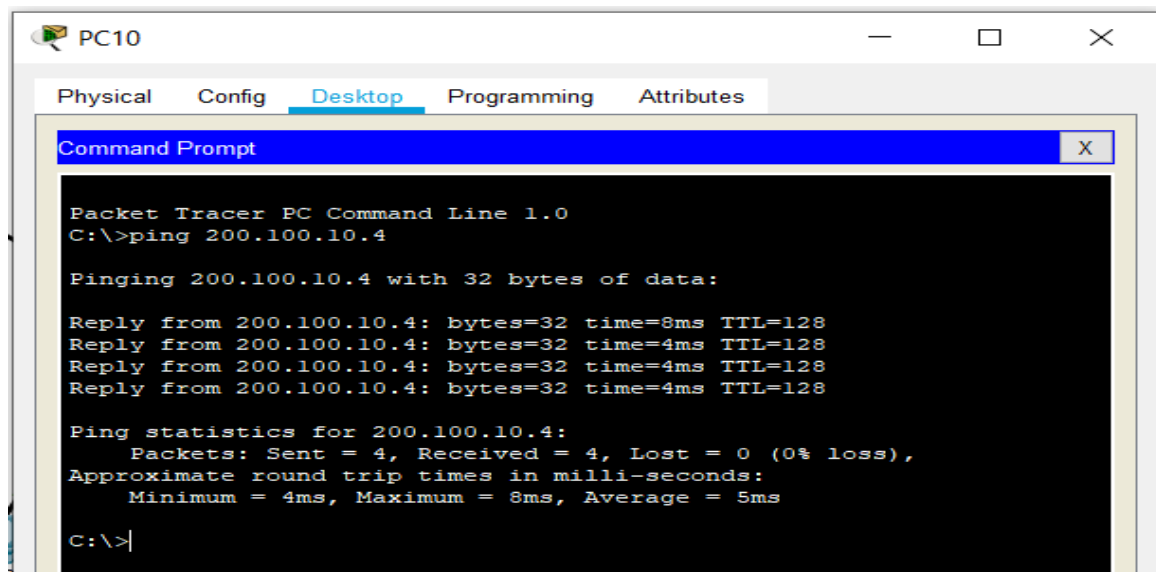
Number of IP Address per subnets: 128

Subnet	Network ID	Broadcast ID	First Host	Last Host
1	200.100.10.0/25	200.100.10.127/25	200.100.10.1/25	200.100.10.126/25
2	200.100.10.128/25	192.168.1.255/26	200.100.10.129/25	200.100.10.254/25

Configuration:

Device	Interface / Subnet	IP Address	Subnet Mask	Default Gateway
Router0	FastEthernet0/0	200.100.10.1/25	255.255.255.128	N/A
Router0	FastEthernet0/1	200.100.10.129/25	255.255.255.128	N/A
PC0	Subnet 1	200.100.10.2/25	255.255.255.128	200.100.10.1/25
PC1	Subnet 1	200.100.10.3/25	255.255.255.128	200.100.10.1/25
PC2	Subnet 1	200.100.10.4/25	255.255.255.128	200.100.10.1/25
PC3	Subnet 2	200.100.10.130/25	255.255.255.128	200.100.10.129/25
PC4	Subnet 2	200.100.10.131/25	255.255.255.128	200.100.10.129/25
PC5	Subnet 2	200.100.10.132/25	255.255.255.128	200.100.10.129/25

Network Topology:



PC10

Physical Config **Desktop** Programming Attributes

Command Prompt

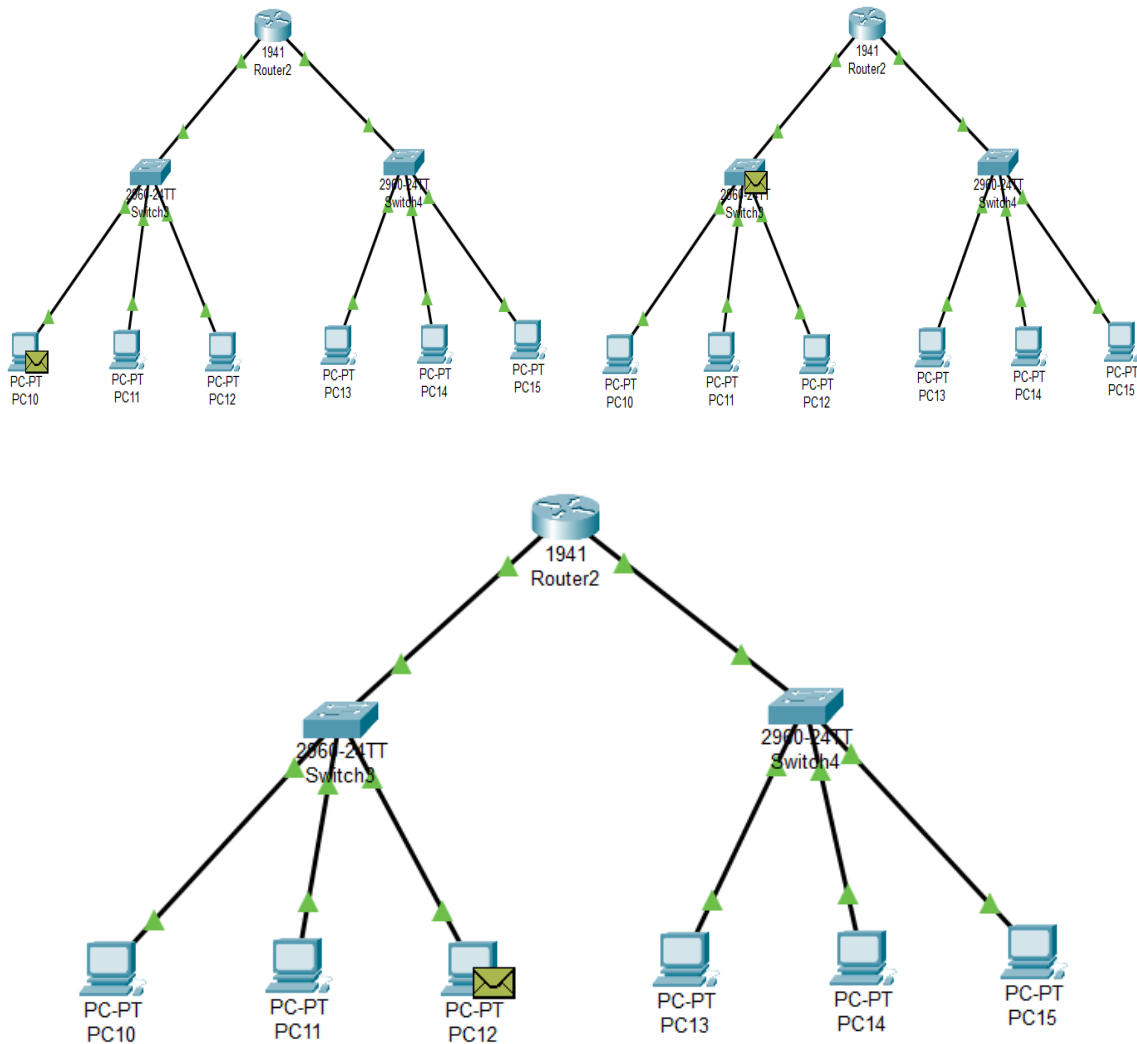
```
Packet Tracer PC Command Line 1.0
C:\>ping 200.100.10.4

Pinging 200.100.10.4 with 32 bytes of data:

Reply from 200.100.10.4: bytes=32 time=8ms TTL=128
Reply from 200.100.10.4: bytes=32 time=4ms TTL=128
Reply from 200.100.10.4: bytes=32 time=4ms TTL=128
Reply from 200.100.10.4: bytes=32 time=4ms TTL=128

Ping statistics for 200.100.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms

C:\>
```



PC12

Physical Config Desktop Programming Attributes

Command Prompt

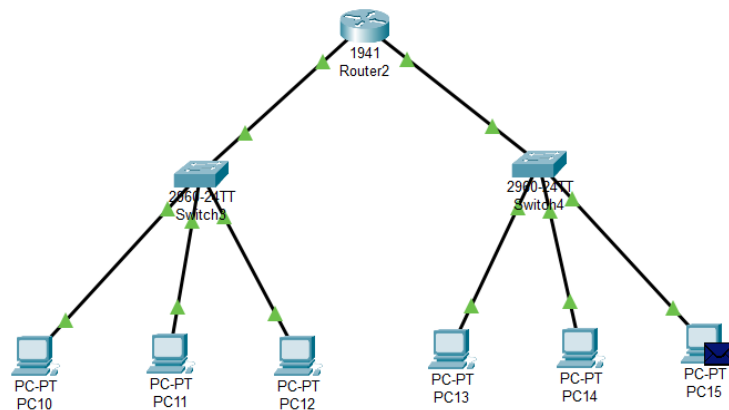
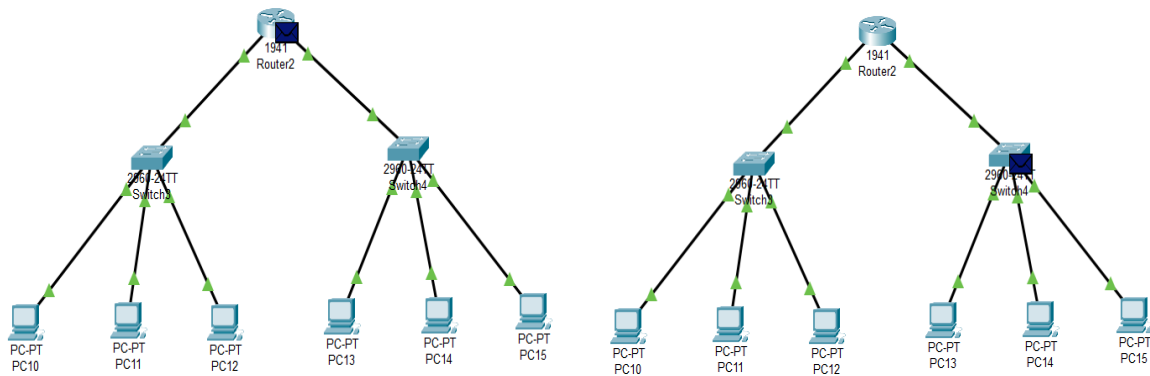
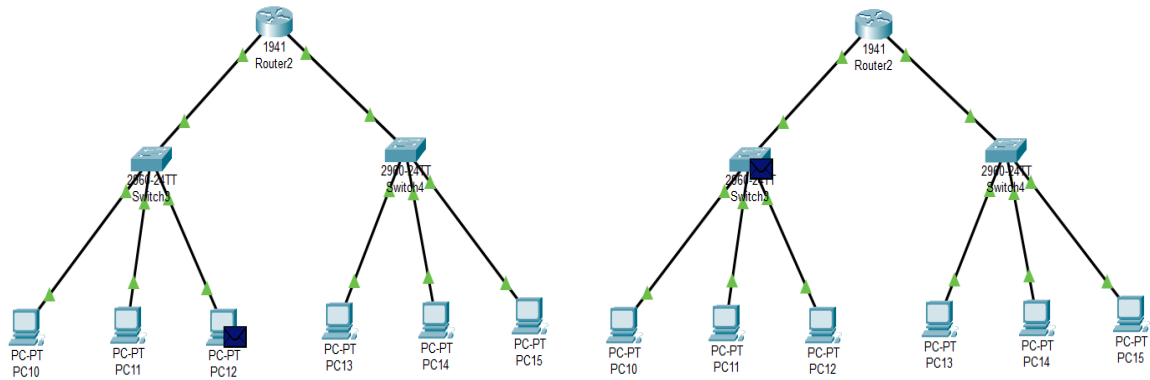
```
Packet Tracer PC Command Line 1.0
C:\>ping 200.100.10.132

Pinging 200.100.10.132 with 32 bytes of data:

Request timed out.
Reply from 200.100.10.132: bytes=32 time=8ms TTL=127
Reply from 200.100.10.132: bytes=32 time=8ms TTL=127
Reply from 200.100.10.132: bytes=32 time=8ms TTL=127

Ping statistics for 200.100.10.132:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 8ms, Average = 8ms

C:\>
```



Result

The lab successfully demonstrated the implementation of both subnetting and supernetting using Cisco Packet Tracer.

Subnetting Results:

The base network 192.168.1.0/24 was divided into 4 subnets, each with 64 usable IP addresses using a /26 subnet mask.

Devices were assigned IP addresses within these subnets. For example, PC0 was configured with 192.168.1.2 in Subnet 1, and PC3 with 192.168.1.66 in Subnet 2.

In assignment work, the 200.100.10.0/24 network split into 2 subnets using a /25 mask, providing 128 addresses per subnet.

Supernetting Results:

Four contiguous networks (192.168.0.0/24 through 192.168.3.0/24) were successfully summarized into a single supernet: 192.168.0.0/22.

The router interface was configured with the supernet mask 255.255.252.0, enabling communication across all networks within the summarized block.

Discussion

This lab focused on the logical manipulation of IP address spaces to meet specific network requirements. Subnetting allowed the division of a large network into smaller, manageable segments by borrowing bits from the host portion of an IP address. This segmentation reduced broadcast traffic and improved overall network performance, while calculations using formulas such as 2^n for subnets and $2^h - 2$ for hosts ensured accurate address allocation. Supernetting, on the other hand, combined multiple contiguous networks into a single routing entry, reducing the size of routing tables. This optimization is particularly useful for ISPs and large-scale networks, as it improves routing efficiency and conserves memory on networking hardware. Additionally, practical tasks such as configuring default gateways in Cisco Packet Tracer demonstrated that subnetting enhances security and management, whereas supernetting supports scalability and simplified routing.

Conclusion

The objectives of this lab were successfully achieved by designing and implementing subnetting and supernetting configurations in Cisco Packet Tracer. The experiments confirmed that subnetting is an effective method for dividing a network into smaller, secure, and high-performing segments, while supernetting optimizes routing by summarizing multiple networks into a single address block. The lab also highlighted the importance of accurately calculating subnet masks and network IDs, as proper planning ensures connectivity and prevents address conflicts. Overall, the exercises reinforced both the theoretical and practical understanding of network addressing and routing strategies.